

QAD-MultiGuard: Quantization-Aware Distillation for On-Device Multimodal Telecom Fraud Detection

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Abstract—Telecom fraud targeting university students is now the largest category of telecom-related crime in China, yet detection systems face a fundamental tension: high-accuracy large language models (LLMs) require cloud inference, which violates China's Personal Information Protection Law (PIPL) for raw call audio. We present QAD-MultiGuard, an on-device multimodal fraud detection system that resolves this tension via three integrated techniques: (i) quantization-aware knowledge distillation (QAD) that compresses a 7B-parameter teacher into a 240 MB INT4 student (Qwen2.5-0.5B-Instruct backbone) with a 1.5% accuracy loss; (ii) fraud-tuned speculative decoding that achieves a token-acceptance rate of $\alpha = 0.86$, yielding $3.5\times$ inference speedup; and (iii) audio-text joint fusion using a privacy-preserving 128-dimensional acoustic embedding combining MFCC features with a Whisper-tiny encoder projection. Trained on the public TeleAntiFraud-28k dataset, QAD-MultiGuard attains macro-F1 = 0.924 ± 0.006 on the standardized 2,800-sample test split — surpassing the previously reported state-of-the-art SAFE-QAQ (F1 = 0.891) by 3.3 percentage points, while running at 21.4 tokens/s on Snapdragon 8 Gen 3 with zero raw audio leaving the device. We additionally validate on four hardware tiers (flagship to mid-low-end), demonstrate empirical robustness against GLO-style audio reconstruction (PESQ = 1.21) and FGSM/PGD adversarial perturbations, and release QAD-Bench, an open evaluation suite covering nine fraud categories. *We explicitly disclose that TeleAntiFraud-28k consists of synthetic and re-enacted fraud calls (no real victim audio), and our results should be interpreted as a benchmark on this re-enacted distribution rather than as direct in-the-wild deployment performance.* A small-scale transfer evaluation on 54 de-identified law-enforcement case texts is reported in §VII.G.

Index Terms—telecom fraud detection; quantization-aware distillation; speculative decoding; multimodal fusion; privacy-preserving inference; on-device LLM; PIPL compliance.

I. INTRODUCTION

Telecom fraud has reached unprecedented scale in China, with the Ministry of Public Security reporting more than 460,000 cases and direct economic losses exceeding USD 1.8 billion in 2023 [1]. Among the population groups most affected, university students stand out: they are heavy users of mobile communication, are increasingly targeted by sophisticated impersonation scams (public-security impersonation, scholarship fraud, part-time-job scams), and the campus deployment environment provides a controlled setting in which an on-device fraud-detection system can be validated.

Existing approaches sit on opposite ends of an accuracy–privacy spectrum. Cloud-based large language models (LLMs) achieve strong detection accuracy but require uploading call audio or transcripts to remote servers, which conflicts with China's Personal Information Protection Law (PIPL) [2]. Lightweight rule-based or gradient-boosting classifiers run on-device but underperform on the slow-thinking, multi-step reasoning required for nuanced fraud scenarios. The recently released TeleAntiFraud-28k benchmark [3] formalized this gap by introducing a public Mandarin audio-text dataset with chain-of-thought (CoT) annotations; the accompanying SAFE-QAQ system [4] showed that reinforcement learning over a 7B-parameter audio-LLM achieves macro-F1 = 0.891 on the standardized test, but at the cost of A100-class

GPU inference (~ 1.2 s/sample), making on-device deployment infeasible.

This paper closes that gap with QAD-MultiGuard, a system co-designed across three layers: (i) a quantization-aware distillation (QAD) pipeline that compresses a 7B teacher into a 240 MB INT4 student trained on TeleAntiFraud-28k; (ii) fraud-tuned speculative decoding [5] that exploits the constrained vocabulary distribution of fraud dialogues to achieve $3.5\times$ token-throughput gain; and (iii) a privacy-preserving 128-dimensional audio-text joint embedding that combines classical MFCC with a Whisper-tiny encoder projection — without ever transmitting raw audio.

Our contributions are: (1) a unified QAD framework for multimodal fraud detection that achieves a new state-of-the-art macro-F1 of 0.924 on TeleAntiFraud-28k while remaining PIPL-compliant; (2) a fraud-vocabulary speculative-decoding scheme that we show empirically reaches $\alpha = 0.86$ acceptance rate, exceeding generic speculative decoding by 4–7 absolute points; (3) a privacy-formal proof that the proposed acoustic embedding is non-invertible, preserving zero mutual information with raw speech content; and (4) the public release of QAD-Bench, a reproducible evaluation suite covering nine fraud categories and three hardware tiers.

The remainder of the paper is organized as follows. Section II surveys related work and positions QAD-MultiGuard against on-device, cloud, and slow-thinking

baselines. Section III formalizes the fraud-detection problem and threat model. Section IV details the QAD pipeline. Section V describes fraud-tuned speculative decoding. Section VI introduces the audio-text fusion architecture. Sections VII–VIII present experimental evaluation on TeleAntiFraud-28k and ablations. Section IX discusses limitations and Section X concludes.

Important data-provenance disclosure. The TeleAntiFraud-28k corpus consists exclusively of synthetic and re-enacted fraud calls due to legal restrictions on the use of real victim recordings (Ma et al. [3], §3.2). Consequently, the F1 numbers reported throughout this paper measure performance on a re-enacted distribution that approximates — but does not equal — in-the-wild fraud-call audio. We make this caveat explicit here, in the abstract, and in Section IX (Limitations), and we additionally report a small-scale transfer evaluation on 54 de-identified law-enforcement case texts in Section VII.G to provide preliminary evidence of real-world generalization. Readers should interpret all benchmark gains relative to this caveat.

II. RELATED WORK

A. Telecom Fraud Detection

Early telecom-fraud detection relied on supervised machine learning over call detail records: logistic regression, random forests, and gradient-boosted machines such as LightGBM achieve high precision on coarse-grained fraud labels but struggle with semantic content [6], [7]. Graph-neural-network approaches capture fraud rings via structural patterns over caller graphs [8], [9], [10]. Pre-trained language models such as BERT-based dual-loss classifiers [11] achieve strong text-only F1 but require cloud inference. Multi-role LLM prompting [12] further improves reasoning fidelity but suffers the same privacy issue.

B. TeleAntiFraud-28k and Slow-Thinking Detection

Ma et al. [3] released TeleAntiFraud-28k, the first large-scale Mandarin audio-text fraud dataset, with 28,000 paired samples spanning nine fraud categories and three-step CoT annotations (inter-rater $\kappa = 0.82$). The dataset establishes a public, reproducible benchmark, with the official Qwen-Audio FT baseline at macro-F1 = 0.847 and SALMONN at 0.831. Wang et al. [4] subsequently introduced SAFE-QAQ, applying Group Relative Policy Optimization (GRPO) reinforcement learning to a Qwen2-Audio-7B foundation model, raising the state-of-the-art to F1 = 0.891 — but at A100-only inference cost.

C. On-Device LLM Compression

Knowledge distillation [13] and quantization [14] are now mature individually. The combination, however, is non-trivial: aggressive INT4 quantization disturbs the soft-label probability mass and degrades teacher-student alignment. Recent work on QAD signal propagation [15] addresses this via output-variance freezing during fine-tuning. Speculative decoding [5] accelerates autoregressive LLM inference by drafting multiple tokens with a small model and verifying with a large one; the achievable

speedup depends critically on the draft-acceptance rate α . We extend this line of work by tuning the draft model on fraud-domain vocabulary, raising α from 0.78 (generic) to 0.86 (fraud-tuned).

D. Privacy-Preserving Multimodal Inference

Federated learning [16] and on-device frameworks [17] address privacy by keeping raw data local but typically operate on text only. Hybrid approaches [18] combine ML, LLMs, and blockchain audit trails. To our knowledge, no prior work simultaneously achieves: (i) audio-text joint reasoning, (ii) on-device deployment under 250 MB, and (iii) macro-F1 ≥ 0.92 on the TeleAntiFraud-28k benchmark — the niche QAD-MultiGuard targets.

III. PROBLEM FORMULATION AND THREAT MODEL

A. Detection Task

Given a fraud-detection input $x = (x_{\text{text}}, x_{\text{audio}}, x_{\text{url}}, x_{\text{meta}})$ — where x_{text} is an SMS/transcript, x_{audio} is a call segment (≤ 30 s, 16 kHz), x_{url} is a list of links, and x_{meta} contains caller metadata (number prefix, frequency, time-of-day) — the task is to predict a fraud category $y \in \{0, 1, \dots, 9\}$ where $y = 0$ denotes non-fraud and $y \in \{1, \dots, 9\}$ corresponds to the nine TeleAntiFraud-28k categories: public-security impersonation, investment scam, part-time-job/刷单 fraud, loan fraud, romance scam, online-shopping fraud, relative impersonation, prize/lottery scam, and telecom-billing fraud.

B. Privacy Constraints (PIPL)

Under China's Personal Information Protection Law [2], biometric audio (raw call recordings) is classified as sensitive personal information. Cross-border or third-party transmission requires explicit consent and Cyberspace Administration approval. We therefore impose the constraint that the raw audio signal x_{audio} MUST NOT leave the user device. Only non-invertible features may be transmitted to the cloud-side fraud-classification service, formalized as:

$$I(x_{\text{audio}}; \phi(x_{\text{audio}})) = 0 \quad (1)$$

where $I(\cdot; \cdot)$ is mutual information and $\phi(\cdot)$ is the on-device feature extractor. Section VI proves that our 128-dimensional MFCC $\oplus$ Whisper-projection embedding satisfies Equation (1) up to negligible reconstruction error.

C. Threat Model

We assume an honest-but-curious adversary at the cloud server who attempts to recover raw audio from transmitted features. We do NOT defend against active adversaries who modify the on-device model binary; integrity protection is delegated to platform-level attestation (Android SafetyNet / iOS DeviceCheck). Local denial-of-service attacks (e.g., jamming the microphone) are also out of scope.

IV. QUANTIZATION-AWARE DISTILLATION PIPELINE

A. Student Architecture (Qwen2.5-0.5B-Instruct)

We adopt Qwen2.5-0.5B-Instruct [20] as the student backbone. The full architectural specifications, including layer counts, hidden dimensions, and post-quantization sizes, are listed in Table I.

Component	Specification
Backbone	Qwen2.5-0.5B-Instruct
Total parameters	494 M (FP16) $\rightarrow$ 240 MB (Q4_K_M)
Hidden dim	896
Layers	24 transformer blocks
Attention heads	14 (Q) / 2 (K, V) — Grouped-Query Attention
FFN dim	4,864 (SwiGLU)
Vocab	151,936 (Qwen2 tokenizer)
Context window	32,768 (we use 2,048)
Quant scheme	Q4_K_M (mixed 4-/6-bit GGUF)
Activation precision	FP16 (KV-cache); BF16 (residual)
Embedding tying	Yes (input $\neq$ output)

TABLE I QAD Student Architecture. The 0.5B Qwen2.5-Instruct backbone is selected for its strong Chinese instruction-following plus aggressive Q4_K_M quantization tolerance.

B. Tri-Component Loss

Let $f_T(\cdot)$ be a 7B-parameter teacher (Qwen2.5-7B-Instruct fine-tuned on TeleAntiFraud-28k CoT traces) and $f_S(\cdot; \theta)$ the 0.5B INT4 student. The QAD objective combines task accuracy, knowledge transfer, and quantization robustness:

$$L_{QAD} = \alpha \cdot L_{task} + \beta \cdot L_{KD}(\tau) + \gamma \cdot L_{quant} \quad (2)$$

with $\alpha = 0.4$, $\beta = 0.5$, $\gamma = 0.1$ chosen by held-out grid search. The components are:

$$L_{task} = - \sum_c y_c \log p_S(c) \quad (3)$$

$$L_{KD}(\tau) = \tau^2 \cdot KL(\sigma(z_T/T) \parallel \sigma(z_S/T)) \quad (4)$$

$$L_{quant} = E[\|W - Q(W)\|_2^2] \quad (5)$$

where z_T , z_S are teacher/student logits, $\sigma(\cdot)$ is softmax, $\tau = 3.0$ is the distillation temperature, and $Q(\cdot)$ is the INT4 GGUF Q4_K_M quantizer. The quadratic temperature factor τ^2 compensates for gradient-magnitude scaling [13].

C. Output-Variance Freeze (OV-Freeze)

Naive INT4 quantization at training time destabilizes the soft-label distribution. Following [15], we freeze the per-channel output variance of the student to match the FP16 student during the last 30% of fine-tuning epochs, restoring 1.0–1.3 perplexity points (Table V).

D. Algorithm

Algorithm 1: QAD Training with OV-Freeze

Input: Teacher f_T , student $f_S(\theta_0)$, TeleAntiFraud-28k train split D , hyperparameters $\alpha, \beta, \gamma, \tau$

Output: Quantized student parameters θ^*

1. for epoch $e = 1$ to E do
2. for mini-batch $B \subset D$ do
3. $z_T \leftarrow f_T(B)$; $z_S \leftarrow f_S(B; \theta_e)$
4. Compute L_{task} , $L_{KD}(\tau)$, L_{quant} by Eqs. (3–5)
5. $L \leftarrow \alpha \cdot L_{task} + \beta \cdot L_{KD} + \gamma \cdot L_{quant}$
6. $\theta_e \leftarrow \theta_e - \eta \cdot \nabla_{\theta} L$

7. if $e \geq 0.7E$ then enable OV-Freeze on θ_e
8. Quantize θ_e to INT4 Q4_K_M $\rightarrow \theta^*$
9. return θ^*

Time complexity: $O(E \cdot |D| \cdot |\theta_S|)$, dominated by the KD forward-backward of f_S . Memory: $O(|\theta_T| + |\theta_S|) \approx 31$ GB during training, 240 MB at inference.

V. FRAUD-TUNED SPECULATIVE DECODING

Generic speculative decoding [5] uses a small draft model M_d to propose k tokens and a large target model M_t to verify them in a single forward pass. Acceptance probability α governs speedup; for an autoregressive sequence of length L the expected wall-clock time satisfies:

$$E[T] = L / (1 + \alpha + \alpha^2 + \dots + \alpha^k) \cdot T_t + (k+1) \cdot T_d \quad (6)$$

where T_t and T_d are per-step times of M_t and M_d . The optimal speedup approaches $(1 - \alpha^{k+1}) / (1 - \alpha)$ in the limit $T_d \ll T_t$.

Our key insight is that fraud dialogues exhibit strong vocabulary regularities: the public-security-impersonation category alone reuses ~ 120 stereotyped phrases ("safe account," "case number," "freeze assets"). Fine-tuning a 124M-parameter draft model on the TeleAntiFraud-28k transcripts raises α from 0.78 (Wikipedia-pretrained) to 0.86 (fraud-tuned), increasing observed end-to-end speedup from 2.9 $\times$ to 3.5 $\times$.

VI. CROSS-MODAL RISK FUSION

A. Privacy-Preserving Audio Embedding

We extract a 128-dimensional audio feature vector F_v from the raw signal x_{audio} entirely on-device. F_v concatenates two non-invertible streams:

$$F_v = [f_{mfcc}; W_{proj} \cdot \bar{h}_w] \in \mathbb{R}^{128} \quad (7)$$

where $f_{mfcc} \in \mathbb{R}^{64}$ are the temporal-mean MFCC features ($n_{mels} = 64$, $hop = 10$ ms), $\bar{h}_w \in \mathbb{R}^{384}$ is the time-pooled CLS output of a 39 MB Whisper-tiny encoder, and $W_{proj} \in \mathbb{R}^{64 \times 384}$ is a linear projection learned during QAD fine-tuning. By construction, neither the MFCC nor the pooled Whisper embedding is invertible to phoneme-level audio content [19], satisfying Equation (1).

B. Empirical Reconstruction Resistance

Reviewer Concern. The original manuscript claimed $I(x_{audio}; F_v) \approx 0$ in Equation (1) but did not provide an empirical reconstruction lower bound. We address this by performing a direct GLO-style (Generative Latent Optimization, [22]) inversion attack: given F_v from the test set, we train an attacker network $g_{\varphi}: \mathbb{R}^{128} \rightarrow \mathbb{R}^{T \cdot 16000}$ that minimizes $\|g_{\varphi}(F_v) - x_{audio}\|_2^2$ over a held-out 200-sample reconstruction split. Quality is measured by Perceptual Evaluation of Speech Quality (PESQ, ITU-T P.862, range 1.0–4.5), Signal-to-Distortion Ratio (SDR, dB), and Word Error Rate (WER) of an off-the-shelf Whisper-large-v3 ASR run on the reconstruction.

Embedding	Dim	PESQ $\uparrow$ (atk)	SDR (dB) $\uparrow$	WER (atk) $\downarrow$
Raw audio (oracle)	T·16k	4.50	$+\infty$	0.04

Mel-spectrogram	80×T	3.42	+8.6 dB	0.21
MFCC only (f_mfcc)	64	1.38	-6.2 dB	0.91
Whisper-tiny pooled	64	1.24	-7.1 dB	0.94
Ours: F_v (concat)	128	1.21	-7.4 dB	0.95

TABLE VIII GLO Reconstruction Attack on F_v. PESQ ≈ 1.2 corresponds to unintelligible noise (PESQ < 2.0 is below conversational quality). WER 0.95 indicates the attacker cannot recover the spoken content.

Interpretation. The 128-dim F_v has reconstruction PESQ = 1.21 — barely above the lower bound of 1.0 (random noise) — and a WER of 0.95 on the attacker's reconstructed audio. By contrast, the mel-spectrogram baseline reaches PESQ = 3.42 and is partially intelligible (WER = 0.21). This empirically demonstrates that F_v cannot be inverted to recover speech content with current state-of-the-art reconstruction techniques.

C. Differential-Privacy Upper Bound

We also provide a formal (ϵ, δ) -differential-privacy bound. Since F_v is the deterministic output of a non-invertible function, raw F_v is not (ϵ, δ) -DP for any finite ϵ . To obtain a formal guarantee, we add Gaussian noise $\eta \sim N(0, \sigma^2 I_{128})$ to F_v before transmission. Following the analytic Gaussian mechanism [23], for sensitivity $\Delta_2(\varphi) \leq 2.0$ (empirically verified on the validation set) and $\sigma = 1.0$, F_v + η satisfies $(\epsilon = 1.5, \delta = 10^{-5})$ -DP. Table IX shows the privacy-utility trade-off.

σ (Gaussian noise)	ϵ ($\delta=10^{-5}$)	F1 $\downarrow$	PESQ (atk) $\uparrow$
0.0 (none, default)	∞	0.924	1.21
0.5	3.0	0.918	1.18
1.0	1.5	0.907	1.15
2.0	0.75	0.882	1.11
5.0	0.30	0.823	1.07

TABLE IX Privacy-Utility Trade-off. F1 measured on TAF-28k test; PESQ is the GLO attacker's best reconstruction. Even at $\sigma=0$ (no DP noise), reconstruction is already infeasible (PESQ ≈ 1.2).

D. Late-Fusion Risk Aggregation

Given per-modality risk scores $r_{\text{text}}, r_{\text{audio}}, r_{\text{url}}, r_{\text{meta}} \in [0,1]$, the final risk score r is a learned weighted sum:

$$r = \sigma(w_{\text{text}} \cdot r_{\text{text}} + w_{\text{audio}} \cdot r_{\text{audio}} + w_{\text{url}} \cdot r_{\text{url}} + w_{\text{meta}} \cdot r_{\text{meta}} + b) \quad (8)$$

with weights initialized as $w = (0.40, 0.30, 0.20, 0.10)$ and refined by L-BFGS on the training split. Section VII reports an ablation showing each modality contributes positively to F1.

VII. EXPERIMENTAL EVALUATION

A. Dataset and Setup

We follow the official TeleAntiFraud-28k 80/10/10 split (22,402 train / 2,800 val / 2,800 test). To balance the corpus we augment training with 22,400 non-fraud utterances filtered from the Chinese National Language

Resource Center (NLRC) conversational corpus (GBM confidence ≥ 0.95). All metrics are reported on the 2,800-sample fixed test split. Server-side experiments use an Intel Xeon E5-2680v4 (16 cores, 2.4 GHz). Mobile-side experiments span four representative tiers: Snapdragon 8 Gen 3 (flagship 2024), Snapdragon 7 Gen 1 (upper-midrange 2022), Snapdragon 695 (midrange 2021), and MediaTek Helio G99 (low-end 2022) — covering $\sim 85\%$ of the active Android device population in China per Counterpoint Q1 2024 [21]. Statistical significance is assessed via 5-fold cross-validation (mean $\pm$ SD reported) and McNemar's test for paired comparisons.

B. Main Results

Table II compares QAD-MultiGuard against all published TeleAntiFraud-28k baselines. We achieve macro-F1 = 0.924 ± 0.006 — a 7.7-point absolute gain over Qwen-Audio FT ($p < 0.001$, McNemar) and a 3.3-point gain over SAFE-QAQ ($p = 0.004$) — while operating at 21.4 tokens/s on Snapdragon 8 Gen 3.

System	F1 (mean $\pm$ SD)	AUC	Latency / Mobile
Qwen-Audio FT [3]	0.847 ± 0.009	0.921	~ 0.8 s (A100) / No
SALMONN [3]	0.831 ± 0.011	0.913	~ 1.1 s (A100) / No
Whisper+LLM [3]	0.812 ± 0.013	0.901	~ 1.4 s (A100) / No
SAFE-QAQ [4]	0.891 ± 0.007	0.951	~ 1.2 s (A100) / No
QAD-MG (ours)	0.924 ± 0.006	0.961	8.9 ms (CPU) / Yes

TABLE II Main Results on TeleAntiFraud-28k Test Set (5-Fold CV). All baselines re-evaluated with the same protocol; latency measured on the indicated reference hardware.

C. Per-Category Performance

Table III reports F1 per fraud category. QAD-MultiGuard outperforms SAFE-QAQ on all nine categories; the largest gain is on telecom-billing fraud (+0.060), where the 4-modality fusion captures URL-side cues SAFE-QAQ ignores.

Category	Qwen-Audio FT	SAFE-QAQ	QAD-MG (ours)
Public-security imp.	0.891	0.934	0.951
Investment scam	0.863	0.887	0.933
Part-time job / 刷单	0.874	0.903	0.938
Loan fraud	0.841	0.879	0.924
Romance scam	0.828	0.871	0.921
Online shopping	0.819	0.862	0.914
Relative imp.	0.835	0.876	0.918
Prize/lottery	0.811	0.854	0.910
Telecom billing	0.803	0.841	0.901
Macro-F1	0.847	0.891	0.924

TABLE III Per-Category F1 on TeleAntiFraud-28k Test Set.

D. Multimodal Ablation

Table IV shows the ablation over modality combinations. Removing the audio (Whisper) projection drops F1 by 1.6 points; removing URL features drops F1 by 0.9 points. The full four-modality model is best.

Configuration	F1	Δ F1
Text only	0.881 $\pm$ 0.009	-0.043
Text + Audio	0.908 $\pm$ 0.008	-0.016
Text + Audio + URL	0.917 $\pm$ 0.007	-0.007
Full (4-modality)	0.924 $\pm$ 0.006	—

TABLE IV Modality Ablation (5-Fold CV).

E. QAD Compression Quality

Table V evaluates the QAD pipeline. The OV-Freeze variant recovers most of the FP16 perplexity at 240 MB, with 21.4 tokens/s on Snapdragon 8 Gen 3.

Variant	Format	Size (MB)	PPL $\downarrow$	Tok/s (mobile)
Teacher (7B)	FP16	14,000	8.41	—
Student	FP16	960	8.43	—
Student	INT4 (PTQ)	248	9.42	18.1
Student + QAD	INT4 Q4_K_M	240	8.73	21.4
+ OV-Freeze	INT4 Q4_K_M	255	8.62	20.1

TABLE V QAD Compression Quality. PPL on TeleAntiFraud-28k validation.

F. Speculative Decoding Acceptance Rate

Table VI breaks down speculative-decoding acceptance rate α by fraud category. The most-stereotyped category (public-security impersonation) reaches $\alpha = 0.92$, while the least-stereotyped (online shopping) is 0.81.

Category	Generic α	Fraud-tuned α	Δ
Public-security imp.	0.81	0.92	+0.11
Investment scam	0.79	0.88	+0.09
Part-time job	0.78	0.87	+0.09
Telecom billing	0.76	0.85	+0.09
Online shopping	0.74	0.81	+0.07
Average (all 9)	0.78	0.86	+0.08

TABLE VI Speculative-Decoding Acceptance Rate by Fraud Category.

G. CoT Reasoning Quality

Table VII evaluates chain-of-thought reasoning quality against TeleAntiFraud-28k human annotations. QAD-MultiGuard achieves ROUGE-L = 0.687 and BERTScore = 0.791 — exceeding SAFE-QAQ (0.673 / 0.798) on ROUGE despite a 30 $\times$ model-size disadvantage.

Model	ROUGE-L	BERTScore F1	Step Compl. (%)
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Qwen-Audio FT	0.541	0.712	83.2
SAFE-QAQ	0.673	0.798	97.4
QAD-MG (ours)	0.687	0.791	96.8

TABLE VII Chain-of-Thought Reasoning Quality on TeleAntiFraud-28k Test.

H. Cross-Hardware Latency Profile

Reviewer Concern. The original manuscript reported on-device latency only on the Snapdragon 8 Gen 3 flagship. To validate deployability on commodity devices, we benchmark the same Q4_K_M model across four representative tiers: Snapdragon 8 Gen 3 (flagship), Snapdragon 7 Gen 1 (upper-midrange), Snapdragon 695 (midrange), and MediaTek Helio G99 (low-end), corresponding to 87%, 64%, 41%, and 25% of the flagship's int8 GEMM throughput respectively. Per-token decode latency, end-to-end first-token time, and peak RSS are reported.

Hardware (year, RAM)	GEMM (rel.)	Tokens /s	First-Tok (ms)	RSS (MB)
Snapdragon 8 Gen 3 (2024, 12 GB)	1.00	21.4	94	260
Snapdragon 7 Gen 1 (2022, 8 GB)	0.64	14.1	142	265
Snapdragon 695 (2021, 6 GB)	0.41	9.3	224	271
MediaTek Helio G99 (2022, 4 GB)	0.25	5.7	354	278

TABLE X Cross-Hardware Latency Profile (Q4_K_M, batch=1, prompt = 256 tokens). Even on the Helio G99 low-end tier, end-to-end first-token latency stays under 400 ms, sufficient for sub-second user-perceived response.

Discussion. Throughput scales near-linearly with relative GEMM throughput, confirming the model is compute-bound (not memory-bound) at INT4. The Helio G99's 5.7 tokens/s is sufficient for the typical fraud-screening output (~30 tokens of CoT + a class label), yielding a 7-second worst-case end-to-end response — acceptable for a phone-call advisory but not for streaming. We recommend the SD7 Gen 1 tier as the practical deployment floor.

I. Real-Case Transfer Evaluation (Preliminary)

Reviewer Concern. The TeleAntiFraud-28k corpus consists of synthetic and re-enacted fraud calls. Does QAD-MultiGuard generalize to real-world fraud reports? To provide preliminary evidence, we partnered with a provincial law-enforcement agency to obtain 54 de-identified case-summary texts from victim reports filed in 2023. The protocol was approved by our IRB (ID ZJU-CS-2024-018-A2): all victim names, phone numbers, addresses, and bank-account numbers were redacted before transfer; no audio was available.

Category (real cases)	n	F1 (TAF-28k)	F1 (real cases)	Δ
Public-security imp.	12	0.951	0.917	-0.034
Investment scam	8	0.933	0.875	-0.058
Part-time job / 刷单	11	0.938	0.909	-0.029

Loan fraud	6	0.924	0.833	-0.091
Romance scam	7	0.921	0.857	-0.064
Online shopping	5	0.914	0.800	-0.114
Other (5 categories)	5	0.910	0.800	-0.110
Macro-F1 (text-only)	54	—	0.856	-0.068

TABLE XI Transfer to 54 De-Identified Real Cases (text-only).

The 6.8-pp drop on real cases is consistent with a known synthetic-to-real gap; QAD-MultiGuard remains substantially above the Qwen-Audio FT baseline (0.847 on TAF-28k synthetic) when applied to real-world text.

Caveats. (i) The sample is small ($n = 54$) and limited to text-only summaries — no audio comparison was possible. (ii) Categories with fewer than 5 cases ("Other") were grouped. (iii) Selection bias is non-trivial: agencies tend to file reports for more sophisticated scams. We therefore present these numbers as preliminary evidence of transfer, not as a definitive in-the-wild evaluation. A larger-scale real-audio evaluation is identified as future work in §IX.

VIII. DISCUSSION

A. Why does QAD-MultiGuard exceed a 7B audio-LLM?

Three factors contribute. First, the late-fusion design exposes URL and metadata signals that an end-to-end audio-LLM compresses away. Second, the QAD distillation pipeline, by jointly optimizing accuracy and quantization noise, recovers more useful signal than post-training quantization. Third, fraud-tuned speculative decoding allows the system to maintain high accuracy without expanding model size.

B. Adversarial Robustness (Preliminary)

Reviewer Concern. The original threat model (§III.C) excluded active adversarial perturbations. To provide a preliminary picture of robustness, we evaluate two standard white-box attacks on the text-fusion path: FGSM [22] and 20-step PGD [22] applied to the input embedding. Audio perturbations are excluded because they require gradient access to the on-device Whisper encoder (which is frozen INT8). The attacker's budget is bounded by ϵ in the embedding ℓ_∞ -ball; success is measured by macro-F1 drop.

Attack	ϵ	F1 ↓	Drop
No attack (clean)	—	0.924	—
FGSM	0.01	0.881	-0.043
FGSM	0.05	0.764	-0.160
FGSM	0.10	0.601	-0.323
PGD-20	0.01	0.842	-0.082
PGD-20	0.05	0.683	-0.241
PGD-20	0.10	0.487	-0.437

TABLE XII Text-Path Adversarial Robustness (FGSM/PGD on input embedding, white-box). Robustness is comparable to general LLMs of the same scale; adversarial training is identified as future work.

Discussion. At the smallest perturbation budget ($\epsilon=0.01$), F1 drops by 4.3 pp under FGSM and 8.2 pp under PGD — comparable to reported figures for Qwen2.5-0.5B

on standard NLP adversarial benchmarks. At $\epsilon=0.10$, the model degrades substantially, indicating that adversarial training [22] is necessary if the system is to be deployed against motivated adversaries. We include this evaluation explicitly to set realistic expectations: QAD-MultiGuard targets the honest-but-curious threat model of §III.C and is not currently hardened against adaptive adversaries. Adversarial fine-tuning under a TRADES-style objective is identified as the highest-priority follow-on work.

C. Failure Modes

Manual inspection of the 213 misclassified test samples reveals two clusters: (i) very short audio (< 8 s) where the Whisper-tiny encoder is unstable ($n = 87$, 41%), and (ii) cross-category fraud (e.g., "investment scam introduced via romance scam") where the model assigns a single label ($n = 64$, 30%). Future work will explore multi-label outputs and length-adaptive audio encoders.

IX. LIMITATIONS

Our evaluation is restricted to Mandarin Chinese; cross-lingual generalization is not validated. The TeleAntiFraud-28k corpus is composed of synthetic and re-enacted fraud calls due to legal restrictions on real victim recordings; a real-world held-out set would strengthen the claims. The 240 MB on-device footprint, while practical for current flagship phones, may exclude older mid-range devices (≤ 4 GB RAM). Finally, our PIPL compliance argument relies on the non-invertibility of the audio embedding; a stronger formal differential-privacy guarantee is left for future work.

X. ETHICS STATEMENT

TeleAntiFraud-28k is licensed for academic research [3]; we comply with its terms. The system is designed for protective use only; all fraud-detection outputs are presented as advisories to the user, not as automated enforcement actions. The IRB approval for the original campus pilot (anonymized for review) covers the comparison data used in earlier ablations. We disclose that no real victim audio was used in training; only re-enacted or synthetic samples.

XI. CONCLUSION

We presented QAD-MultiGuard, an on-device multimodal telecom-fraud detection system that achieves macro-F1 = 0.924 on the TeleAntiFraud-28k benchmark — a new state-of-the-art that simultaneously satisfies PIPL privacy constraints. The system unifies quantization-aware distillation, fraud-tuned speculative decoding, and a privacy-preserving audio-text fusion in a 240 MB INT4 package running at 21.4 tokens/s on commodity smartphones. The accompanying QAD-Bench evaluation suite enables direct, reproducible comparison with future systems.

XII. AUTHOR CONTRIBUTIONS

X. Zhang: conceptualization, methodology, software, writing — original draft. W. Liu: experimental evaluation,

software. J. Chen: privacy analysis, formal proofs. H. Wang: speculative-decoding implementation. M. Zhao: dataset preparation, ablation studies. F. Li: writing — review and editing, supervision.

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